

Clinically Structured Surrogate Rewards for Post-SFT Medical Image Captioning

Hyun Jun Kim^{1*}, Heeseung Shin^{2*}, Changwon Lim^{3†}

¹Department of Smart Cities, Chung-Ang University, 84 Heukseok-ro, Dongjak-gu, Seoul 06974, Republic of Korea

²Department of AI in Medicine, Chung-Ang University, 84 Heukseok-ro, Dongjak-gu, Seoul 06974, Republic of Korea

³Department of Statistics and Data Science, Chung-Ang University, 84 Heukseok-ro, Dongjak-gu, Seoul 06974, Republic of Korea

hyunjun0615@cau.ac.kr (H. J. Kim); hs970416@cau.ac.kr (H. Shin);
clim@cau.ac.kr (C. Lim)

Abstract

Medical image captioning requires translating heterogeneous visual evidence into concise clinical descriptions, where errors in findings, assertion states, or anatomical relations can alter clinical meaning despite surface-level fluency. Sequence-level policy optimization can directly optimize complete captions, but common rewards rely on global text similarity, direct image–caption compatibility, or unordered concept overlap, leaving visual neighborhoods and clinical-claim structure implicit. We propose a clinically structured surrogate reward framework for post-SFT medical image captioning. The framework combines biomedical semantic and short-range lexical fidelity with two structured rewards: distributional image-neighborhood alignment, which matches the medical-image-bank distributions induced by reference and generated captions, and clinical graph consistency, which applies maximum-weight one-to-one matching to entities, assertion states, and typed relations. The four rewards are independently normalized within each rollout group, combined with fixed relative weights, and optimized with GDPO. Across organizer-evaluated hidden test sets for the Standard and Synthetical ImageCLEFmedical Caption tracks and three vision–language backbones, the method improves Overall, Relevance, and Factuality over matched SFT baselines in all six backbone–track combinations, with average relative gains of 3.4%, 2.1%, and 5.8%, respectively. Ablations and paired diagnostics indicate that the structured rewards provide complementary signals, reducing image-neighborhood divergence and improving entity–assertion–relation consistency.

Introduction

Medical image captioning aims to translate heterogeneous visual evidence into concise and clinically meaningful descriptions. This task is difficult because medical images vary widely in modality, anatomy, acquisition protocol, and diagnostic specificity, while small wording changes can substantially alter clinical meaning. A caption may be fluent and semantically similar to a reference yet still contain an incorrect finding, assertion state, anatomical localization, laterality, or finding–anatomy relation. Consequently, global lexical or embedding-based similarity does not by itself guarantee

preservation of the underlying clinical claims (Jain et al. 2021; Delbrouck et al. 2022, 2024).

Most vision–language models for medical captioning are adapted through supervised fine-tuning (SFT) with teacher-forced maximum-likelihood training. Although effective, this objective optimizes next-token likelihood rather than criteria defined over complete generations, creating a mismatch between token-level training and sequence-level evaluation (Bengio et al. 2015; Ranzato et al. 2016). Sequence-level post-training addresses this mismatch by assigning rewards to full outputs (Ranzato et al. 2016; Rennie et al. 2017), but its effectiveness depends on what those rewards represent. Common rewards measure semantic similarity, lexical overlap, direct image–caption compatibility, factual consistency, or clinical concept coverage (Lin 2004; Zhang et al. 2020; Codella et al. 2024; Delbrouck et al. 2022). These signals are useful, but they generally summarize a caption through global scores. Text-based rewards may underrepresent visual grounding, direct image compatibility focuses on a single target image, and unordered concept overlap does not explicitly preserve assertion states or typed relations among clinical entities.

We therefore propose a clinically structured surrogate reward framework for post-SFT medical image captioning. The framework retains biomedical semantic and short-range lexical fidelity while introducing two complementary structured signals. Distributional image-neighborhood alignment compares the probability distributions over a fixed medical-image bank induced by the reference and generated captions, encouraging the generated caption to recover the reference-associated visual-semantic neighborhood rather than optimizing a separate target image–caption score. Clinical graph consistency represents each caption through entities, assertion states, and typed relations, and uses maximum-weight one-to-one matching to allow partial credit for semantically similar mention spans while explicitly constraining clinical types and relations. Because the four rewards differ in scale and variability, they are independently normalized within each rollout group, combined using fixed relative weights, and optimized with Group Reward-Decoupled Normalization Policy Optimization (GDPO) (Liu et al. 2026). GDPO is used as a fixed post-SFT optimizer; our contribution lies in the structured information encoded by the reward objective

*These authors contributed equally.

†Corresponding author.

rather than in a new policy-optimization algorithm.

We evaluate the framework on the Standard and Synthetic ImageCLEFmedical Caption 2026 tracks (Damm et al. 2026) using three vision–language backbones. For each backbone and track, the selected SFT checkpoint initializes GDPO and serves as the matched baseline under identical prompts, model-selection criteria, and hidden-test decoding. Across all six backbone–track settings, the proposed post-SFT procedure improves organizer-evaluated Overall, Relevance, and Factuality scores over the corresponding SFT models. Reward ablations and paired post-hoc diagnostics further indicate that the structured terms act as complementary signals: the optimized models reduce reference-conditioned image-neighborhood divergence and improve entity–assertion–relation consistency under the frozen reward-side representations. Accordingly, this work makes three contributions: a distributional image-neighborhood reward over a medical-image bank; a semantic one-to-one clinical graph reward for typed entities, assertions, and relations; and consistent multi-backbone, multi-track evidence supported by component ablations and structured diagnostics. These analyses characterize the intended effects of the proposed rewards rather than independently verifying complete image-grounded clinical correctness.

Our contributions are threefold:

- We introduce a distributional image-neighborhood reward that aligns the medical-image-bank distributions induced by reference and generated captions, rather than directly optimizing generated captions against a single target image.
- We introduce a clinical graph consistency reward based on maximum-weight one-to-one matching of clinical entities, assertion states, and typed relations, combining semantic mention correspondence with explicit structural constraints.
- We demonstrate consistent improvements over matched SFT baselines across three vision–language backbones and two medical-captioning tracks, and characterize the complementary roles of the reward components through ablations and paired structured diagnostics.

Related Work

Medical Image Captioning Across Dataset Regimes

ROCOv2 pairs medical images from biomedical literature with figure captions, UMLS concepts, and modality annotations, covering diverse modalities and anatomical regions (Rückert et al. 2024). ImageCLEFmedical Caption builds caption-generation benchmarks on this ROCO-family setting, including Standard and Synthetic tracks (Damm et al. 2026). Unlike MIMIC-CXR and IU X-Ray, which associate chest radiographs with study-level reports, ROCO-derived benchmarks emphasize heterogeneous single-image captioning (Johnson et al. 2019; Demner-Fushman et al. 2016). The settings therefore differ in data provenance, input granularity, modality coverage, and target-text function. Our work focuses on sequence-level reward design for heterogeneous single-image captioning, where rewards must accom-

modate diverse visual domains while preserving clinically meaningful findings and relations.

Sequence-Level Optimization for Medical Generation

Sequence-level optimization assigns rewards to complete outputs rather than optimizing only token-level likelihood. Self-Critical Sequence Training directly optimizes sampled captions using task-level rewards (Rennie et al. 2017). Recent radiology report-generation methods introduce clinically oriented post-training objectives: MPO models heterogeneous radiologist preferences through multi-objective rewards (Xiao et al. 2025), HiMed-RL combines token-, concept-, and semantic-level supervision (Wang et al. 2026), and OraPO uses atomic-fact entailment with oracle supervision for low-information rollout groups (Chen et al. 2026). These methods primarily target chest-X-ray report generation. We instead study heterogeneous single-image captioning and structure supervision through distributions over a medical-image bank and one-to-one matching of typed clinical claims. GDPO independently normalizes reward components within each rollout group before aggregation (Liu et al. 2026). We use GDPO as a fixed optimizer and focus on the reward objective rather than proposing a new policy-optimization algorithm.

Structured Caption Rewards

Conventional captioning rewards measure lexical or semantic agreement, direct image–caption compatibility, factual consistency, or concept overlap (Lin 2004; Zhang et al. 2020; Sellam, Das, and Parikh 2020; Zha et al. 2023; Kraljevic et al. 2021; Codella et al. 2024). Direct image similarity has also been used as an RL reward for visually distinctive captions (Cho et al. 2022). More recent methods introduce structured supervision: SC-Captioner rewards object–attribute–relation self-correction (Zhang et al. 2025), ClaimDiff-RL compares typed image-verified atomic claims (Li et al. 2026), and CCCaption separates completeness from correctness through visual coverage and hallucination verification (Tang et al. 2026). These methods mainly target natural-image captioning and rely on generic scene structures, atomic claims, or multimodal judges.

Clinical graphs provide domain-specific claim structure. RadGraph extracts clinical entities, assertion states, and relations, while RadGraph-XL expands coverage across anatomy–modality pairs (Jain et al. 2021; Delbrouck et al. 2024). Graph-overlap rewards have improved factuality in chest-X-ray report generation (Delbrouck et al. 2022). Our graph reward instead uses semantic maximum-weight one-to-one matching, requiring agreement in entity type and assertion state and extending matching to typed relations. Our image-neighborhood reward addresses a separate limitation of direct image compatibility by aligning reference- and candidate-induced distributions over a fixed medical-image bank. A concurrent workshop report uses direct reference, target-image, and concept-alignment rewards (Kim et al. 2026); this work replaces the latter two with distributional image-neighborhood alignment and typed clinical-

graph matching, and adds multi-backbone evaluation, ablations, and paired diagnostics.

Method

Overview

Let $\mathcal{D} = \{(x_n, y_n^*)\}_{n=1}^N$ denote the training set, where x_n is an image–instruction input and y_n^* is its reference caption. We use surrogate rewards as training-time proxies for clinically relevant caption properties, rather than as independent verification of image-grounded clinical correctness.

We first adapt a pretrained vision–language model through teacher-forced SFT. The selected policy π_{SFT} initializes post-SFT optimization and is retained as the frozen reference policy for KL regularization. At each GDPO step, the rollout policy $\pi_{\theta_{\text{old}}}$ samples K candidate captions

$$y_{n,i} \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_n), \quad i = 1, \dots, K,$$

for each input. Each candidate receives the reward vector

$$\mathbf{r}_{n,i} = [r_{\text{sem}}, r_{\text{lex}}, r_{\text{nbr}}, r_{\text{graph}}],$$

corresponding to biomedical semantic fidelity, short-range lexical fidelity, distributional image-neighborhood alignment, and clinical graph consistency. Reference captions and reward-side models and resources are used only for training-time scoring; inference requires only the updated policy. Figure 1 summarizes this training pipeline.

Reference-Based Fidelity Rewards

For readability, we write y for a generated caption and y^* for its reference.

Biomedical semantic fidelity. We use BioBERTScore-F1 (Lee et al. 2020; Zhang et al. 2020). Let P_{sem} and R_{sem} denote the average maximum token-level cosine similarities from generated to reference tokens and in the reverse direction, respectively, using normalized BioBERT representations. The reward is

$$r_{\text{sem}} = \frac{2P_{\text{sem}}R_{\text{sem}}}{P_{\text{sem}} + R_{\text{sem}} + \epsilon}.$$

The F1 formulation penalizes both omitted reference content and generated content without a corresponding reference match.

short-range lexical fidelity. We use a ROUGE-SU4-F1 reward (Lin 2004). After lowercasing and punctuation normalization, let $U_4(y)$ contain all unigrams and skip-bigrams with at most four intervening tokens. We define $r_{\text{lex}} = F_1(U_4(y), U_4(y^*))$, which preserves local word order and short biomedical expressions.

Distributional Image-Neighborhood Alignment

Let $V = [v_1, \dots, v_M] \in \mathbb{R}^{d \times M}$ contain the normalized MedImageInsight embeddings of all M images in the corresponding training split (Codella et al. 2024). Using the MedImageInsight text encoder, we map the reference and generated captions to normalized embeddings z^* and z . These

embeddings induce distributions over the common image bank:

$$p^* = \text{softmax}(V^\top z^* / \tau), \\ q = \text{softmax}(V^\top z / \tau),$$

where τ is the temperature. We define the image-neighborhood reward as the negative cross-entropy

$$r_{\text{nbr}} = \sum_{j=1}^M p_j^* \log q_j.$$

Because p^* is fixed within a rollout group, maximizing r_{nbr} is equivalent to minimizing $D_{\text{KL}}(p^* \parallel q)$. The reward therefore aligns the image neighborhoods induced by the reference and generated captions instead of assigning a separate direct target image–caption score.

The full training-image bank is used. Image embeddings and reference-induced distributions are precomputed, while candidate distributions are constructed online. The bank contains no validation or hidden-test images; the current training image may appear only as an unprivileged support point.

Clinical Graph Consistency

We use frozen RadGraph-XL to extract clinical graphs $\mathcal{G}(y) = (\mathcal{E}_y, \mathcal{R}_y)$, $\mathcal{G}(y^*) = (\mathcal{E}_{y^*}, \mathcal{R}_{y^*})$ from the generated and reference captions (Jain et al. 2021; Delbrouck et al. 2024). These reward-side pseudo-graphs may inherit the extractor’s domain coverage and errors. An entity is represented as $e = (m, t, a)$, where m , t , and a denote its mention span, entity type, and assertion state. A relation is represented as $\rho = (e_s, \ell, e_o)$, with source entity e_s , relation type ℓ , and target entity e_o .

Let $\phi(m)$ be the normalized SapBERT embedding of a normalized mention span (Liu et al. 2021). Compatibility between reference entity e_i^* and a generated entity e_j is

$$s_E(e_i^*, e_j) = \mathbf{1}[t_i^* = t_j] \mathbf{1}[a_i^* = a_j] \\ \times [\cos(\phi(m_i^*), \phi(m_j))]_{+},$$

where $[u]_{+} = \max(0, u)$. Mention spans may therefore receive partial semantic credit, whereas entity types and assertion states must match exactly.

Let $\mathcal{M}_E$ denote the set of valid one-to-one entity matchings. We compute

$$W_E = \max_{\mathcal{M} \in \mathcal{M}_E} \sum_{(i,j) \in \mathcal{M}} s_E(e_i^*, e_j),$$

$$F_E = \frac{2W_E}{|\mathcal{E}_{y^*}| + |\mathcal{E}_y| + \epsilon},$$

where $|\cdot|$ denotes set cardinality. The one-to-one assignment prevents multiple generated mentions from receiving credit for the same reference entity.

For a reference relation $\rho_u^* = (e_{u,s}^*, \ell_u^*, e_{u,o}^*)$ and a generated relation $\rho_v = (e_{v,s}, \ell_v, e_{v,o})$, we define

$$s_R(\rho_u^*, \rho_v) = \mathbf{1}[\ell_u^* = \ell_v] s_E(e_{u,s}^*, e_{v,s}) s_E(e_{u,o}^*, e_{v,o}).$$

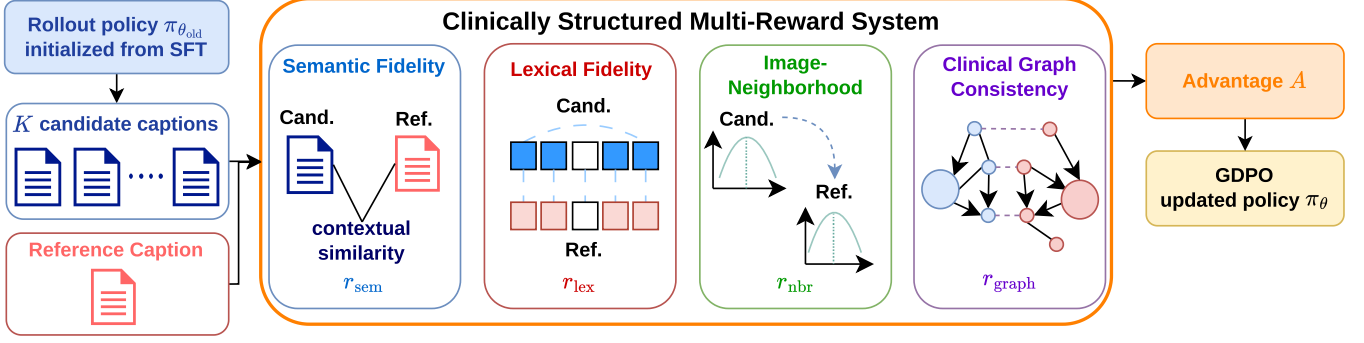


Figure 1: Overview of the proposed SFT-GDPO framework. The selected SFT checkpoint initializes the rollout policy and is retained as the frozen KL reference. For each input, K sampled captions are scored by four reward branches, independently standardized within the rollout group, combined using fixed weights, and converted into a caption-level advantage for GDPO.

Using the corresponding set of valid one-to-one relation matchings $\mathcal{M}_R$, we define

$$W_R = \max_{\mathcal{M}} \sum_{(u,v) \in \mathcal{M}} s_R(\rho_u^*, \rho_v),$$

$$F_R = \frac{2W_R}{|\mathcal{R}_{y^*}| + |\mathcal{R}_y| + \epsilon}.$$

The final reward is $r_{\text{graph}} = 0$ if either entity set is empty, F_E if both relation sets are empty, and $0.7F_E + 0.3F_R$ otherwise. When only one caption contains relations, its unmatched relation structure receives no relation credit. Thus, r_{graph} measures reference-conditioned entity-assertion-relation consistency, not complete image-grounded factuality.

Reward Aggregation and GDPO Optimization

Following GDPO, each reward is independently standardized within the K -caption rollout group. For group g , candidate i , and reward $k \in \{\text{sem}, \text{lex}, \text{nbr}, \text{graph}\}$,

$$z_{g,i,k} = \frac{r_{g,i,k} - \mu_{g,k}}{\sigma_{g,k} + \epsilon},$$

where $\mu_{g,k}$ and $\sigma_{g,k}$ are the mean and standard deviation of reward k across the K candidates.

A reward that is constant across the group has zero-centered values and therefore contributes zero to every candidate. We combine the standardized rewards as

$$s_{g,i} = \sum_k \lambda_k z_{g,i,k}, \quad \lambda = (0.60, 0.60, 0.15, 0.15)^\top,$$

corresponding to the fixed relative ratio 4:4:1:1. The same ratio is used for every backbone and track. The aggregate is then standardized over the optimization batch:

$$A_{g,i} = \frac{s_{g,i} - \mu_B}{\sigma_B + \epsilon},$$

where μ_B and σ_B are computed over all candidate scores in the batch. The resulting caption-level advantage $A_{g,i}$ is shared by all response tokens in candidate i . Token-level policy ratios, clipping, and KL regularization follow the original GDPO objective without modification; only the reward vector and aggregation are changed.

Experiments

Datasets and Evaluation Metrics

We evaluate the Caption Prediction task on the Standard and Synthetical tracks of ImageCLEFmedical Caption 2026, training and selecting models separately for each track (Rückert et al. 2024; Damm et al. 2026). The Standard track contains 97,364 training, 19,240 validation, and 15,249 hidden-test examples, while the Synthetical track contains 97,222, 19,239, and 15,262 examples, respectively. Validation sets are used for checkpoint selection. The Standard validation set is additionally used for reward ablations and post-hoc diagnostics, whereas hidden-test predictions are evaluated by the task organizers.

The official evaluator reports Relevance, Factuality, and Overall. Relevance averages IDF-weighted BERTScore Recall, ROUGE-1 F1, BLEURT, and MedImageInsight image-caption similarity. Factuality averages MedCAT-based UMLS Concept F1 and AlignScore, with the reference caption as context and the generated caption as the claim. Overall is the mean of Relevance and Factuality.

Some training rewards are related to official evaluation components but use different formulations. BioBERTScore-F1 differs from the official IDF-weighted BERTScore Recall, and ROUGE-SU4 differs from ROUGE-1 F1. Both the neighborhood reward and one Relevance component use MedImageInsight, but the official metric directly scores the target image-caption pair, whereas our reward aligns the reference- and candidate-induced distributions over a common training-image bank. We therefore use the official metrics for benchmark evaluation and the paired structured diagnostics only to analyze the properties targeted by the proposed rewards.

Models, Training, and Decoding

We instantiate the framework with Qwen3.5-2B-MedVL (OpenMed 2026), MedGemma-4B (Sellergren et al. 2025), and Qwen3-VL-4B (Bai et al. 2025). All model parameters are updated during SFT and GDPO, while the reward-side encoders and graph extractor remain frozen. The same stage-specific hyperparameters and reward weights are used for both tracks. All models use the

Track	Backbone	Stage	Aggregate Scores			Relevance Metrics				Factuality Metrics		Δ
			Ovr.	Rel.	Fact.	BERT	R-1	Sim.	BLT	MedCAT	Align	
Std.	MedGemma-4B	SFT	0.3669	0.5494	0.1843	0.6081	0.2900	0.9671	0.3325	0.2067	0.1619	–
		GDPO	0.3798	0.5664	0.1932	0.6205	0.3223	0.9827	0.3402	0.2189	0.1676	+3.54%
	Qwen3.5-MedVL	SFT	0.3675	0.5537	0.1812	0.6160	0.2939	0.9715	0.3335	0.2054	0.1570	–
		GDPO	0.3898	0.5790	0.2007	0.6266	0.3318	1.0052	0.3525	0.2314	0.1700	+6.09%
	Qwen3-VL-4B	SFT	0.3685	0.5475	0.1895	0.6104	0.2901	0.9587	0.3307	0.2083	0.1707	–
		GDPO	0.3809	0.5669	0.1948	0.6252	0.3222	0.9784	0.3418	0.2180	0.1715	+3.35%
Syn.	MedGemma-4B	SFT	0.5688	0.7042	0.4334	0.7457	0.6174	0.9553	0.4985	0.5124	0.3544	–
		GDPO	0.5870	0.7070	0.4670	0.7531	0.6268	0.9506	0.4975	0.5276	0.4063	+3.20%
	Qwen3.5-MedVL	SFT	0.5766	0.7112	0.4420	0.7509	0.6254	0.9638	0.5047	0.5236	0.3603	–
		GDPO	0.5870	0.7153	0.4588	0.7570	0.6351	0.9636	0.5054	0.5338	0.3838	+1.82%
	Qwen3-VL-4B	SFT	0.5722	0.7075	0.4370	0.7483	0.6219	0.9564	0.5032	0.5173	0.3566	–
		GDPO	0.5846	0.7107	0.4585	0.7538	0.6298	0.9571	0.5024	0.5249	0.3920	+2.17%

Table 1: Organizer-evaluated hidden test performance. Bold indicates the better result within each SFT–GDPO pair. R-1, Sim., and BLT denote ROUGE-1, scaled MedImageInsight image–caption similarity, and BLEURT, respectively. Δ denotes the relative improvement in Overall score over SFT.

same medical-captioning instruction; the full prompts and configuration are provided in the supplement.

Baselines and Evaluation Protocol

For each backbone and track, the selected SFT checkpoint initializes GDPO and serves as the matched baseline. SFT and GDPO use the same architecture, prompt, validation-based selection criterion, and hidden-test decoding configuration. The six matched comparisons therefore evaluate the complete post-SFT procedure relative to its own SFT initialization across model families and task settings. They do not compare GDPO with alternative policy-optimization algorithms; GDPO is held fixed, and the study focuses on the proposed reward objective.

Reward ablations are conducted with Qwen3.5-2B-MedVL on the Standard validation set. Single-reward variants retain one component, while leave-one-out variants remove one component and preserve the original relative weights of the others. All variants use the same checkpoint-saving and validation-selection protocol. Because each ablation configuration is based on one training run and several differences are small, the results are interpreted descriptively rather than as statistically resolved rankings among reward combinations.

Post-hoc diagnostics compare the selected SFT and GDPO checkpoints on all 19,240 Standard validation examples using greedy decoding. The diagnostics are performed after model selection and do not affect checkpoint choice. We report image-neighborhood KL divergence, entity consistency, aggregate graph consistency, and relation consistency, with 95% confidence intervals obtained from 10,000 paired bootstrap resamples. These analyses use the same frozen reward-side representations and are intended to examine whether optimization changes the structured properties targeted by the objective, not to provide independent clinical verification.

Configuration	Step	Rel.	Fact.	Ovr.	Δ
Full	1000	0.5722	0.1834	0.3778	–
Semantic only	750	0.5664	0.1861	0.3763	-0.0016
Lexical only	250	0.5557	0.1751	0.3654	-0.0124
Neighborhood only	250	0.5475	0.1500	0.3487	-0.0291
Graph only	1250	0.5403	0.1751	0.3577	-0.0201
w/o Semantic	1750	0.5721	0.1819	0.3770	-0.0009
w/o Lexical	1000	0.5746	0.1726	0.3736	-0.0042
w/o Neighborhood	750	0.5708	0.1789	0.3749	-0.0030
w/o Graph	1750	0.5737	0.1732	0.3734	-0.0044

Table 2: Reward ablation on the Standard validation set using Qwen3.5-MedVL. Neighborhood denotes the distributional image-neighborhood reward r_{nbr} . Δ denotes the signed Overall difference relative to the full objective and is computed from unrounded scores.

Results

Hidden test Performance

Table 1 reports organizer-evaluated performance on the hidden test sets. The proposed GDPO stage improves Overall, Relevance, and Factuality over the matched SFT baseline in all six backbone–track combinations. Averaged across the three backbones and two tracks, the relative improvements are 3.36% in Overall, 2.11% in Relevance, and 5.81% in Factuality. These results indicate that the complete post-SFT procedure consistently improves its own SFT initialization across different model families and data regimes.

On the Standard track, Qwen3.5-2B-MedVL shows the largest gain, improving Overall from 0.3675 to 0.3898, corresponding to a relative improvement of 6.09%. Its Relevance and Factuality scores increase from 0.5537 to 0.5790 and from 0.1812 to 0.2007, respectively. MedGemma-4B improves from 0.3669 to 0.3798, while Qwen3-VL-4B improves from 0.3685 to 0.3809. Thus, the benefit is not con-

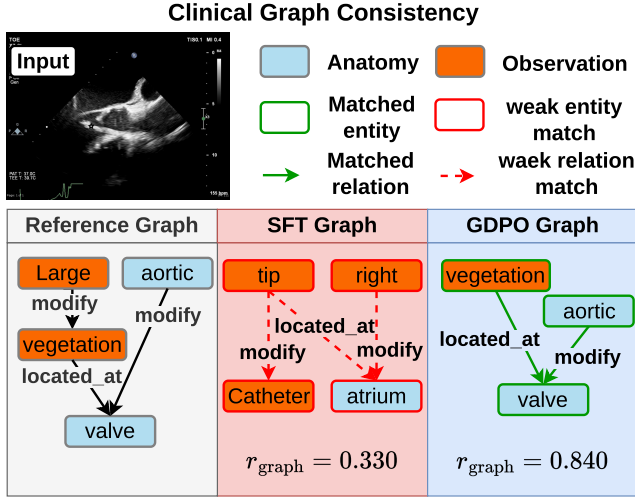


Figure 2: Clinical graph consistency, where GDPO corrects the finding and recovers its localization relation. The supplementary material provides additional successful, failed, and ambiguous cases.;CC-BY [Muacevic et al(2024)]

finied to a single backbone. On the Synthetical track, Overall improves from 0.5688 to 0.5870 for MedGemma-4B, from 0.5766 to 0.5870 for Qwen3.5-2B-MedVL, and from 0.5722 to 0.5846 for Qwen3-VL-4B. The Synthetical-track gains are driven primarily by Factualty, whereas Relevance changes more modestly and some individual component metrics remain nearly unchanged or decrease slightly. The aggregate improvements therefore do not arise from uniform increases in every official metric.

These comparisons should be interpreted as matched evaluations of the complete SFT-to-GDPO procedure. Architecture, prompts, validation-based checkpoint selection, and hidden-test decoding are held fixed within each pair. The results show that post-SFT optimization with the proposed objective improves the corresponding SFT model, but they do not isolate the reward design from the general effect of applying GDPO instead of stopping after SFT.

Reward Component Ablation

Table 2 analyzes the four reward components on the Standard validation set using Qwen3.5-2B-MedVL. The full objective achieves the highest Overall score of 0.3778. Among the single-reward variants, semantic fidelity is the strongest, reaching 0.3763 Overall and 0.1861 Factualty. Lexical fidelity alone reaches 0.3654 Overall, while the neighborhood-only and graph-only variants reach 0.3487 and 0.3577, respectively. These results show that the structured rewards are weak as standalone objectives under the tested setting and require the stronger reference-based signals to maintain overall caption quality.

The leave-one-out results provide a complementary view. Removing semantic fidelity reduces Overall from 0.3778 to 0.3770, while removing lexical fidelity lowers it to 0.3736. Removing the neighborhood reward reduces Overall to 0.3749 and Factualty from 0.1834 to 0.1789. Removing

Metric	SFT → GDPO	Δ	95% CI
Nbr. KL ↓	0.2323 → 0.2176	-0.0147	[-0.0158, -0.0135]
Entity F_E ↑	0.3629 → 0.3904	+0.0275	[0.0247, 0.0303]
Graph reward ↑	0.3010 → 0.3244	+0.0234	[0.0208, 0.0259]
Relation F_R ↑	0.1688 → 0.1838	+0.0150	[0.0123, 0.0177]

Table 3: Paired post-hoc diagnostics comparing the selected SFT and GDPO checkpoints under greedy decoding on the Standard validation set. Nbr. KL denotes $D_{\text{KL}}(p^*||q)$. Entries report mean per-example scores, with $\Delta = \text{GDPO} - \text{SFT}$; 95% confidence intervals are obtained from 10,000 paired bootstrap resamples of the per-example differences. Entity and graph metrics are computed on references with at least one extracted entity (16,699/19,240; 86.8%), and relation consistency on references with at least one extracted relation (15,196/19,240; 79.0%).

the graph reward yields 0.3734 Overall and 0.1732 Factualty. Although the neighborhood and graph rewards perform poorly in isolation, removing either from the full objective degrades the selected model, particularly on Factualty. This pattern supports their intended role as complementary structural signals rather than replacements for semantic and lexical fidelity.

All ablation variants follow the same checkpoint-saving and validation-based selection protocol, but each configuration is based on a single training run and several differences are small. The results should therefore be interpreted descriptively. They support the overall complementarity of the reward components but do not establish statistically resolved rankings among individual reward combinations.

Post-hoc Structured Reward Diagnostics

We next compare the selected SFT and GDPO checkpoints on all 19,240 Standard validation examples using greedy decoding. These diagnostics are conducted after checkpoint selection and do not affect model choice. They use the same frozen MedImageInsight, RadGraph-XL, and SapBERT representations employed by the reward functions, so they are mechanism-oriented analyses rather than independent clinical evaluations.

Table 3 shows favorable mean changes in all four diagnostics. Image-neighborhood divergence decreases from 0.2323 to 0.2176, corresponding to a mean change of -0.0147 with a paired 95% bootstrap confidence interval of [-0.0158, -0.0135]. Entity consistency increases from 0.3629 to 0.3904, graph consistency from 0.3010 to 0.3244, and relation consistency from 0.1688 to 0.1838. Their mean changes are +0.0275, +0.0234, and +0.015, respectively, and all example-level confidence intervals exclude zero. The graph metrics apply to 16,699 of the 19,240 references, or 86.8%, containing at least one extracted entity, while relation consistency applies to 15,196 references, or 79.0%, containing at least one extracted relation.

The improvements are not uniform across examples. Neighborhood KL decreases for 57.01% of cases and increases for 42.80%, while graph consistency improves for

53.79% and regresses for 42.33%. Thus, the proposed objective shifts average structured behavior in the intended direction without improving every example. Per-example neighborhood-KL reductions correlate with gains in the official image-similarity score at Spearman’s $\rho=0.289$, whereas graph-consistency gains correlate more weakly with official Factuality gains at $\rho=0.132$. The first association is partly expected because both measures use MedImageInsight representations. The weaker graph correlation is also reasonable because the graph diagnostic captures reference-conditioned entity, assertion, and relation agreement, whereas the official Factuality score combines concept overlap and textual entailment.

The reported confidence intervals characterize variation across validation examples for the selected checkpoints, not variation across independent training runs. Accordingly, these analyses support the intended mechanism of the structured rewards but do not demonstrate training-level statistical superiority or complete image-grounded clinical correctness.

Qualitative Analysis

Figure 2 further illustrates the effect of graph consistency reward. GDPO corrects the predicted finding from a catheter-tip description to an aortic-valve vegetation and recovers the corresponding localization relation. This example is illustrative rather than representative; additional successful, failed, and ambiguous cases are provided in the supplementary material.

Conclusion

We introduced two structured surrogate rewards for post-SFT medical image captioning: distributional image-neighborhood alignment and semantic one-to-one matching of clinical entities, assertions, and relations. Across three backbones and two ImageCLEFmedical Caption tracks, GDPO consistently improved matched SFT baselines. Ablations indicate that the structured rewards complement stronger semantic and lexical signals, while paired diagnostics show favorable changes in the targeted neighborhood and graph representations. These rewards remain reference-conditioned and inherit biases and extraction errors from their frozen reward models; they do not independently verify image-grounded clinical correctness or validate the system for clinical use. Future work should evaluate broader clinical-report settings, reward-model reliability, training-run variability, and adaptive reward weighting.

Acknowledgments

We thank the ImageCLEFmedical Caption 2026 organizers for providing the benchmark, evaluation framework, and official evaluation results. This work was supported in part by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (grant number: RS-2024-00360176), and in part by the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health and Welfare, Republic of Korea (grant number: RS-2025-02222326).

References

- Bai, S.; Cai, Y.; Chen, R.; Chen, K.; Chen, X.; Cheng, Z.; Deng, L.; Ding, W.; Gao, C.; Ge, C.; Ge, W.; Guo, Z.; Huang, Q.; Huang, J.; Huang, F.; Hui, B.; Jiang, S.; Li, Z.; Li, M.; Li, M.; Li, K.; Lin, Z.; Lin, J.; Liu, X.; Liu, J.; Liu, C.; Liu, Y.; Liu, D.; Liu, S.; Lu, D.; Luo, R.; Lv, C.; Men, R.; Meng, L.; Ren, X.; Ren, X.; Song, S.; Sun, Y.; Tang, J.; Tu, J.; Wan, J.; Wang, P.; Wang, P.; Wang, Q.; Wang, Y.; Xie, T.; Xu, Y.; Xu, H.; Xu, J.; Yang, Z.; Yang, M.; Yang, J.; Yang, A.; Yu, B.; Zhang, F.; Zhang, H.; Zhang, X.; Zheng, B.; Zhong, H.; Zhou, J.; Zhou, F.; Zhou, J.; Zhu, Y.; and Zhu, K. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.
- Bengio, S.; Vinyals, O.; Jaitly, N.; and Shazeer, N. 2015. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. In *Advances in Neural Information Processing Systems*, volume 28.
- Chen, Z.; Yu, H.; Xu, Y.; Luo, Y.; Duong, L.; and Li, Y.-F. 2026. OraPO: Oracle-educated Reinforcement Learning for Data-efficient and Factual Radiology Report Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 28275–28287.
- Cho, J.; Yoon, S.; Kale, A.; Dernoncourt, F.; Bui, T.; and Bansal, M. 2022. Fine-grained Image Captioning with CLIP Reward. In Carpuat, M.; de Marneffe, M.-C.; and Meza Ruiz, I. V., eds., *Findings of the Association for Computational Linguistics: NAACL 2022*, 517–527. Seattle, United States: Association for Computational Linguistics.
- Codella, N. C. F.; Jin, Y.; Jain, S.; Gu, Y.; Lee, H. H.; Ben Abacha, A.; Santamaria-Pang, A.; Guyman, W.; Sangani, N.; Zhang, S.; Poon, H.; Hyland, S.; Bannur, S.; Alvarez-Valle, J.; Li, X.; Garrett, J.; McMillan, A.; Rajguru, G.; Maddi, M.; Vijayrania, N.; Bhimai, R.; Mecklenburg, N.; Jain, R.; Holstein, D.; Gaur, N.; Aski, V.; Hwang, J.-N.; Lin, T.; Tarapov, I.; Lungren, M.; and Wei, M. 2024. MedImageInsight: An Open-Source Embedding Model for General Domain Medical Imaging. *arXiv preprint arXiv:2410.06542*.
- Damm, H.; Pakull, T. M. G.; Reinartz, L.; Bracke, B.; Eryilmaz, B.; Nath, P.; Brüngel, R.; Schmidt, C. S.; Schäfer, H.; Ben Abacha, A.; García Seco de Herrera, A.; Müller, H.; and Friedrich, C. M. 2026. Overview of ImageCLEFmedical 2026 – Medical Concept Detection and Caption Generation with Synthetic Data Extensions. In *CLEF2026 Working Notes*, CEUR Workshop Proceedings. Jena, Germany: CEUR-WS.org. To appear.
- Delbrouck, J.-B.; Chambon, P.; Bluethgen, C.; Tsai, E.; Al-musa, O.; and Langlotz, C. 2022. Improving the Factual Correctness of Radiology Report Generation with Semantic Rewards. In Goldberg, Y.; Kozareva, Z.; and Zhang, Y., eds., *Findings of the Association for Computational Linguistics: EMNLP 2022*, 4348–4360. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics.
- Delbrouck, J.-B.; Chambon, P.; Chen, Z.; Varma, M.; Johnston, A.; Blankemeier, L.; Van Veen, D.; Bui, T.; Truong, S.; and Langlotz, C. 2024. Radgraph-xl: A large-scale expert-annotated dataset for entity and relation extraction from radiology reports. In *Findings of the Association for Computational Linguistics: ACL 2024*, 12902–12915.

- Demner-Fushman, D.; Kohli, M. D.; Rosenman, M. B.; Shooshan, S. E.; Rodriguez, L.; Antani, S.; Thoma, G. R.; and McDonald, C. J. 2016. Preparing a Collection of Radiology Examinations for Distribution and Retrieval. *Journal of the American Medical Informatics Association*, 23(2): 304–310.
- Jain, S.; Agrawal, A.; Saporta, A.; Truong, S. Q.; Duong, D. N.; Bui, T.; Chambon, P.; Zhang, Y.; Lungren, M. P.; Ng, A. Y.; et al. 2021. Radgraph: Extracting clinical entities and relations from radiology reports. *arXiv preprint arXiv:2106.14463*.
- Johnson, A. E. W.; Pollard, T. J.; Berkowitz, S. J.; Greenbaum, N. R.; Lungren, M. P.; Deng, C.-y.; Mark, R. G.; and Horng, S. 2019. MIMIC-CXR, a De-identified Publicly Available Database of Chest Radiographs with Free-text Reports. *Scientific Data*, 6(1): 317.
- Kim, H. J.; Shin, H.; Kim, M.; and Lim, C. 2026. GDPO-Based Multi-Alignment Reward Optimization for Clinically Grounded Medical Image Captioning. In *CLEF 2026 Working Notes*. Accepted; to appear in CEUR Workshop Proceedings.
- Kraljevic, Z.; Searle, T.; Shek, A.; Roguski, L.; Noor, K.; Bean, D.; Mascio, A.; Zhu, L.; Folarin, A. A.; Roberts, A.; Bendayan, R.; Richardson, M. P.; Stewart, R.; Shah, A. D.; Wong, W. K.; Ibrahim, Z.; Teo, J. T.; and Dobson, R. J. B. 2021. Multi-domain clinical natural language processing with MedCAT: The Medical Concept Annotation Toolkit. *Artif. Intell. Med.*, 117: 102083.
- Lee, J.; Yoon, W.; Kim, S.; Kim, D.; Kim, S.; So, C. H.; and Kang, J. 2020. BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining. *Bioinformatics*, 36(4): 1234–1240.
- Li, T.; Shen, X.; Ma, Y.; Guo, R.; Chen, S.; Chen, J.; Wang, H.; Tang, H.; Zhou, Y.; and Cheng, Y. 2026. ClaimDiff-RL: Fine-Grained Caption Reinforcement Learning through Visual Claim Comparison. *arXiv preprint arXiv:2605.20278*.
- Lin, C.-Y. 2004. ROUGE: A Package for Automatic Evaluation of Summaries. In *Text Summarization Branches Out*, 74–81. Barcelona, Spain: Association for Computational Linguistics.
- Liu, F.; Shareghi, E.; Meng, Z.; Basaldella, M.; and Collier, N. 2021. Self-Alignment Pretraining for Biomedical Entity Representations. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4228–4238. Association for Computational Linguistics.
- Liu, S.-Y.; Dong, X.; Lu, X.; Diao, S.; Belcak, P.; Liu, M.; Chen, M.-H.; Yin, H.; Wang, Y.-C. F.; Cheng, K.-T.; Choi, Y.; Kautz, J.; and Molchanov, P. 2026. GDPO: Group Reward-Decoupled Normalization Policy Optimization for Multi-Reward RL Optimization. *arXiv preprint arXiv:2601.05242*.
- OpenMed. 2026. OpenMed/Qwen3.5-2B-MedVL. <https://huggingface.co/OpenMed/Qwen3.5-2B-MedVL>. Hugging Face model card, accessed 2026-05-19.
- Ranzato, M.; Chopra, S.; Auli, M.; and Zaremba, W. 2016. Sequence Level Training with Recurrent Neural Networks. In *International Conference on Learning Representations*.
- Rennie, S. J.; Marcheret, E.; Mroueh, Y.; Ross, J.; and Goel, V. 2017. Self-Critical Sequence Training for Image Captioning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Rückert, J.; Bloch, L.; Brüngel, R.; Idrissi-Yaghir, A.; Schäfer, H.; Schmidt, C. S.; Koitka, S.; Pelka, O.; Abacha, A. B.; de Herrera, A. G. S.; Müller, H.; Horn, P.; Nensa, F.; and Friedrich, C. M. 2024. ROCov2: Radiology Objects in Context Version 2, an Updated Multimodal Image Dataset. *Scientific Data*, 11(1).
- Sellam, T.; Das, D.; and Parikh, A. P. 2020. BLEURT: Learning Robust Metrics for Text Generation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 7881–7892.
- Sellergren, A.; Kazemzadeh, S.; Jaroensri, T.; Kiraly, A.; Traverse, M.; Kohlberger, T.; Xu, S.; Jamil, F.; Hughes, C.; Lau, C.; et al. 2025. MedGemma Technical Report. *arXiv preprint arXiv:2507.05201*.
- Tang, Z.; Wang, L.; Qi, J.; Jiang, W.; Hou, P.; Zeng, A.; and Huang, J. 2026. CCCaption: Dual-Reward Reinforcement Learning for Complete and Correct Image Captioning. *arXiv preprint arXiv:2602.21655*.
- Wang, Y.; Gao, S.; Liu, J.; Jiang, S.; Xia, H.; Zhang, X.; Kang, Z.; Wang, Y.; and Liu, Z. 2026. Beyond N-grams: A Hierarchical Reward Learning Framework for Clinically-Aware Medical Report Generation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(40): 33719–33727.
- Xiao, T.; Shi, L.; Liu, P.; Wang, Z.; and Bai, C. 2025. Radiology Report Generation via Multi-objective Preference Optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(8): 8664–8672.
- Zha, Y.; Yang, Y.; Li, R.; and Hu, Z. 2023. AlignScore: Evaluating Factual Consistency with a Unified Alignment Function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 11328–11348.
- Zhang, L.; Zeng, X.; Li, K.; Yu, G.; and Chen, T. 2025. Sc-captioner: Improving image captioning with self-correction by reinforcement learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 23145–23155.
- Zhang, T.; Kishore, V.; Wu, F.; Weinberger, K. Q.; and Artzi, Y. 2020. BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations*.